



TRAINING LAB. 8

TOPOLOGY OPTIMIZATION OF THE HUB CARRIER OF AN ULTRA-EFFICIENT LIGHTWEIGHT BATTERY ELECTRIC VEHICLE

Introduction

The vehicle designed and manufactured by the Shell Eco-marathon¹ team at Politecnico di Milano is called *Asteria*² (Figure 1) and belongs to the urban concept class, which includes vehicles with architecture inspired by small city cars.

The vehicle body is entirely made from carbon-fiber-based composite materials and the body shape has been designed to minimize the aerodynamic drag. The electric drivetrain is powered by a 48 V lithium-ion battery pack. Two 200 W brushed DC motors are connected to the two rear wheels, the overall nominal power of the vehicle is 400 W. The transmission is realized by two gear pairs that connect each motor to the rear wheel. A freewheel located in the wheel hub allows to transmit power only in one direction. The vehicle speed is controlled by modulating the voltage across the motors through a PWM (pulse width modulation) signal.



Figure 1 – Asteria, the urban concept vehicle designed to compete in the Shell Eco Marathon.

¹ Shell Eco marathon is an international engineering competition addressed to teams of students. The aim is to complete a valid race stint using the least amount of energy, within a given time.

² In Greek mythology, Asteria was the sister of one of the brides of Zeus, Leto, mother of Apollon and Artemis. According to the myth, she transformed herself into a quail to save her sister.

During the motion of a road vehicle, there are three main sources of energy dissipation:

- The total mass of the vehicle (inertia);
- The tire rolling resistance;
- The aerodynamic drag.

A sensitivity analysis conducted on models that reproduce the law of motion of vehicles, reveals that mass is the most influential parameter on energy consumption. For the considered vehicle specific data are available (Figure 2):

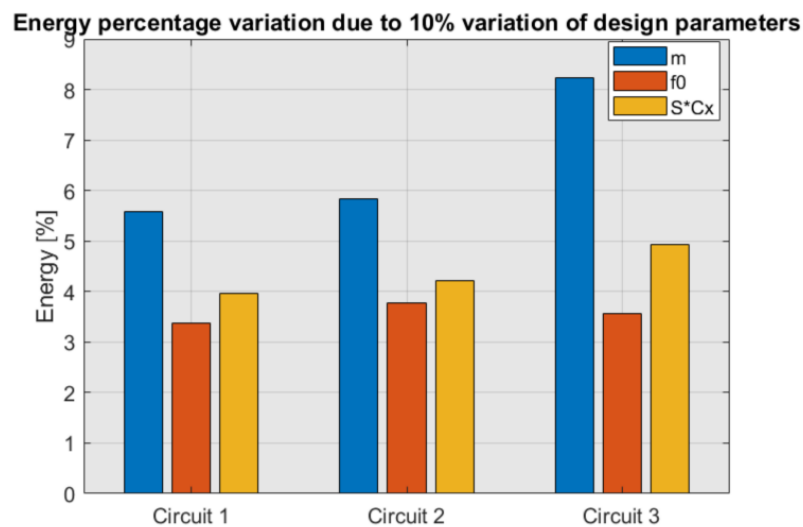


Figure 2 - Sensitivity analysis on energy consumption for three different circuits.

Aim of the project

The team is now working on the design of a new vehicle for competing to the next Shell Eco Marathon. The new car is equipped with a double-wishbone type suspension system like the one shown in *Figure 3*. The aim of the project is to optimize the structural layout of the hub carrier in order to maximize the structural stiffness with the least amount of mass.

The topology optimization problem will be solved by means of the commercial software “Altair Optistruct”.

The hub carrier (shown in *Figure 3*) has three main functions. First, it closes the kinematic loop made by the two control arms; secondly, it receives the inputs from the steering rod and transfers them to the wheel; thirdly, it bears the static and dynamic loads coming from the road and the brake caliper.

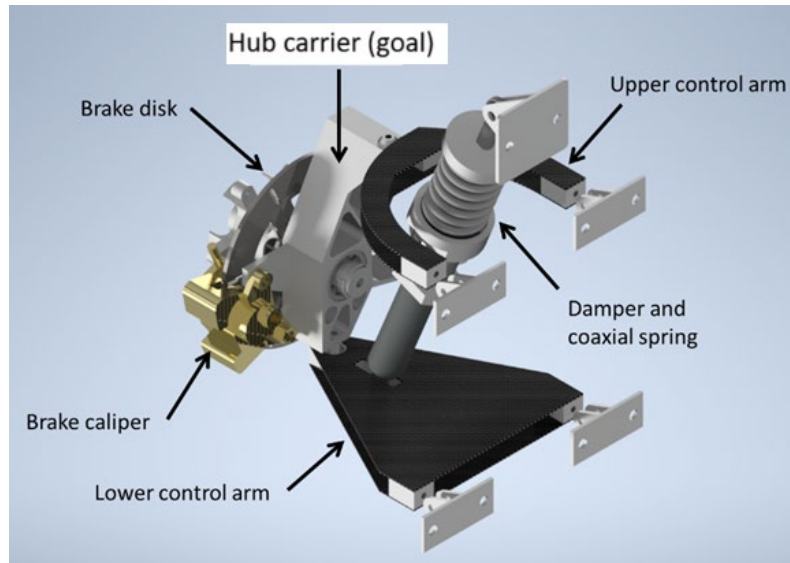


Figure 3 - Front right suspension of the vehicle. The attachment of the steering rod is hidden by the damper.

Figure 4 shows the suspension system from different point of view. The central figure clearly shows the attachment of the steering rod.

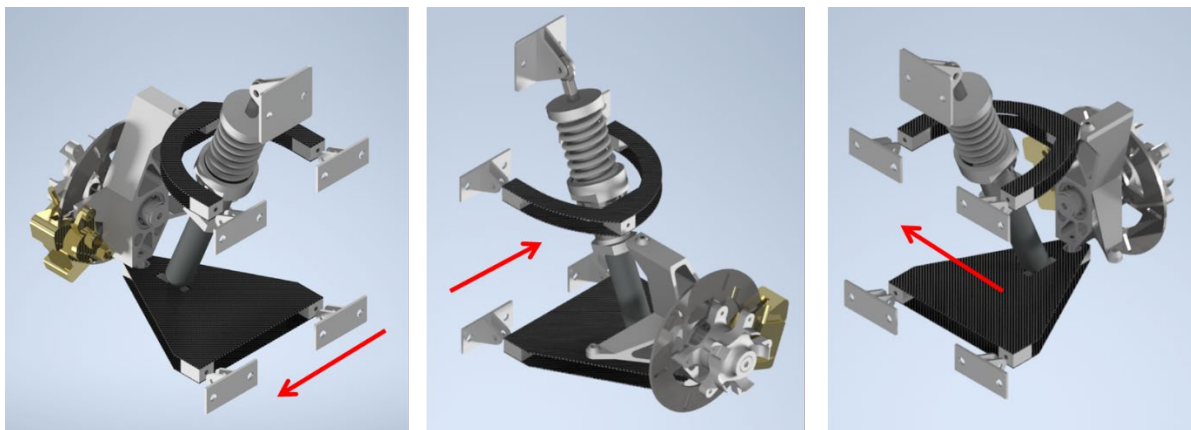


Figure 4 - Assembly of the front right double wishbone suspension of the vehicle. Red arrow pointing toward the front of the car.

Design domain and boundary conditions

Figure 5 shows the design geometry of the hub carrier. It is defined to take into consideration the space available inside the wheel and the position of the holes, which represent functional surfaces (attachments) to be preserved during the iterations. In the figure, red volumes represent portions of the hub carrier that cannot change during the optimization process.

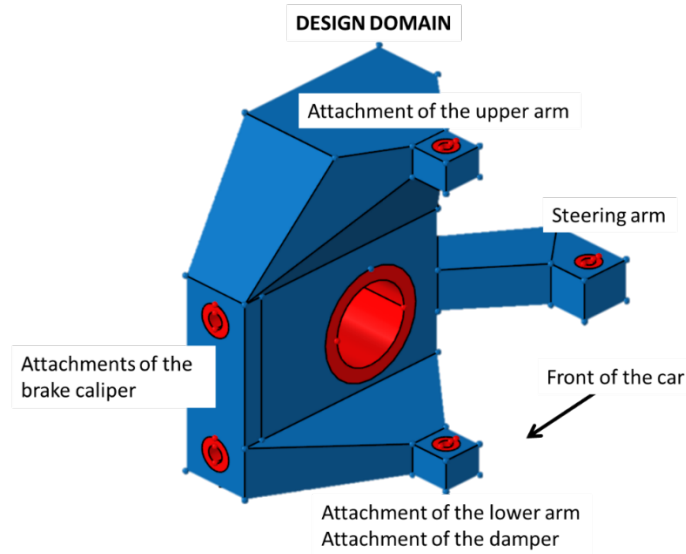


Figure 5 – Design domain of the hub carrier (blue).

At a first approximation, the scheme of Figure 6 can be considered.

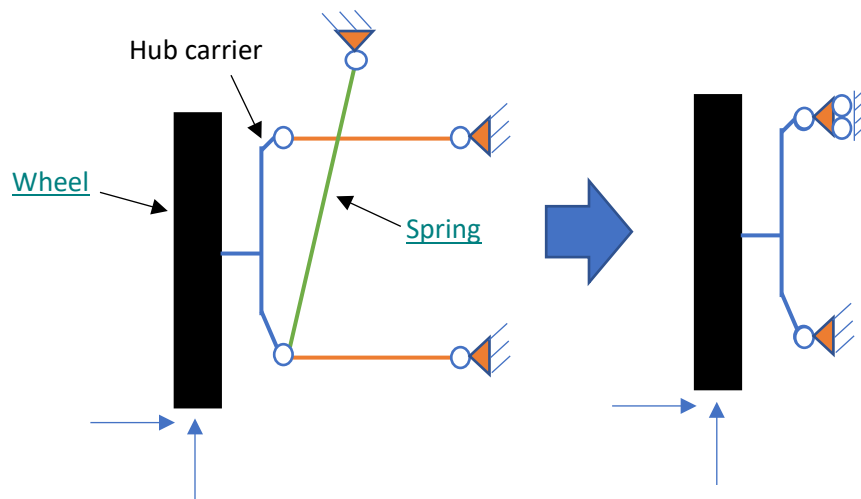


Figure 6 schematics for approximation of boundary conditions.

According to the scheme of Figure 6, the following boundary conditions are imposed to the hub carrier. The displacement along X and Y directions of the upper attachment is constrained, while sliding along Z direction is allowed (Figure 7). For the lower attachment, due to the spring and damper reactions, displacement along X,Y and Z directions is constrained. Lastly the steering arm is linked to the steering rod. The steering mechanism constraints the displacement of the connection point only in the Y direction.

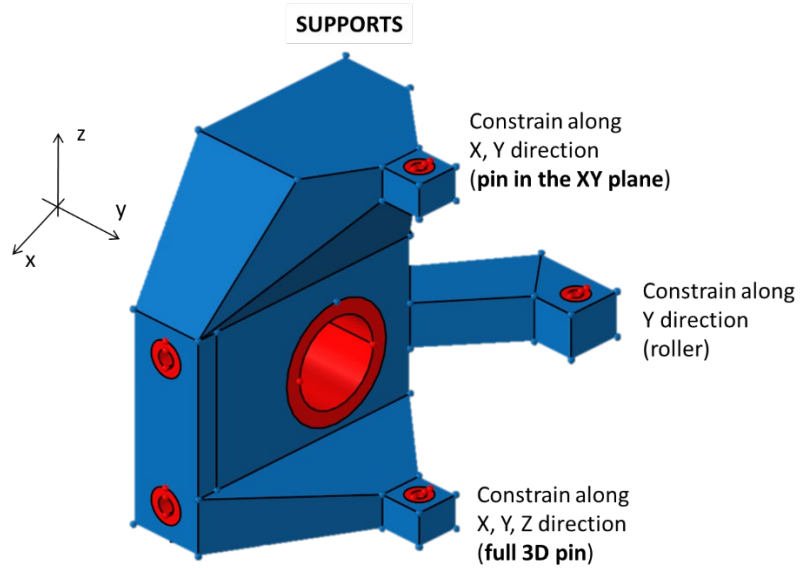


Figure 7 - Constraints.

Load cases

A total of six load cases are considered:

- Pure rolling. The car is coasting, no torques are applied to the wheels;
- Braking. The car is intensively braking;
- Internal turn. The car is steering, the front right wheel is internal to the curve;
- External turn. The car is steering, the front right wheel is external to the curve;
- External obstacle. The car hits a curb from the external side of the front right wheel;
- Internal obstacle. The car hits a curb from the internal side of the front right wheel.

Pure rolling

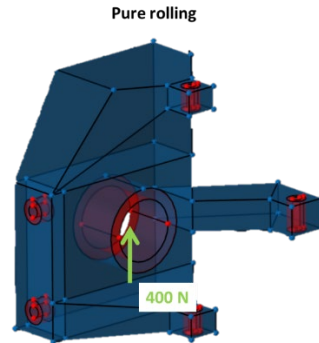


Figure 8 - Pure rolling

In pure rolling condition, the hub carrier sustains just the weight of the car. Considering a mass of 160 kg equally distributed on four wheels, the vertical force acting on each wheel is 400 N.

Braking

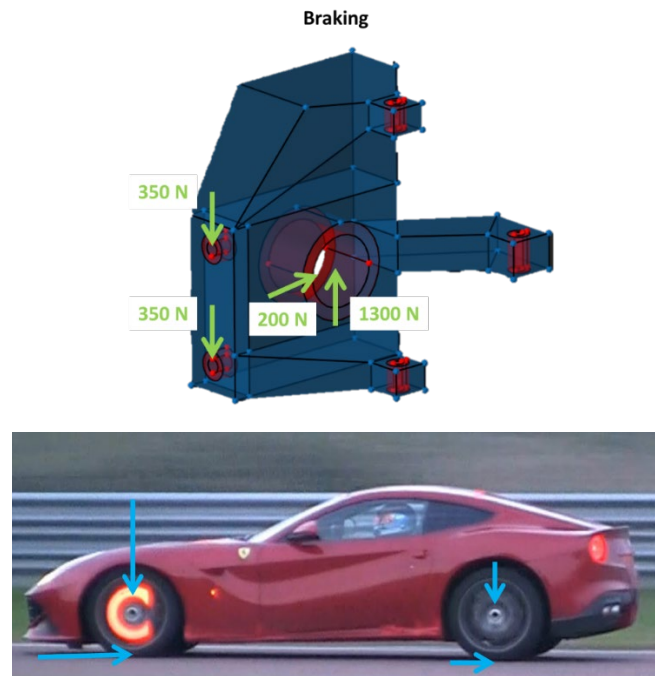


Figure 9 - Braking

During braking, a load transfer occurs from the rear to the front axle. A pitch motion is also observed. The tangential force caused by the friction between disc and pads is applied at the two fixtures of the brake caliper. Such force acting on the caliper is estimated considering a longitudinal braking force of 200 N and the ratio between the wheel and the brake disc radii is 3.5. The tangential force applied at the caliper fixtures (along Z direction) reads

$$F_t = F_{long} \frac{r_{wheel}}{r_{disc}} = 700 \text{ N}$$

The tangential force F_t is equally split on the two fixtures of the caliper as shown in Figure 9.

The total vertical load acting on the center of the hub carrier is 1300 N. This resulting force is given by the following contributions:

- 1) Vertical static load acting on a single wheel equal to 400 N (acting along +Z)
- 2) load transfer contribution in braking of 200 N (acting along +Z)
- 3) A vertical force equal to F_t (700 N) acting at the center of the hub carrier (along +Z) that is balancing the (internal) force acting on the caliper.

Finally, the longitudinal braking force equal to 200 N is applied at the center of the hub carrier (see Figure 9).

Internal turn

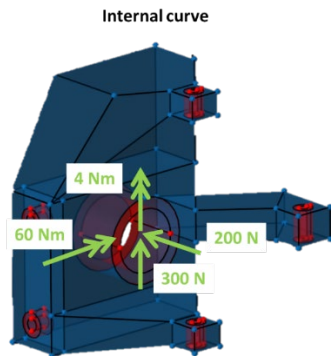


Figure 10 - Internal turn

When a car turns, it must resist centrifugal force. The vertical load is transferred from internal to external wheels. It is assumed that on the internal wheels a vertical load of 300 N is acting (-100 N with respect to the nominal condition). A lateral acceleration of $0.5g$ (5 m/s^2) while cornering is assumed, resulting in a lateral force of 200 N acting on each wheel. The lateral force is applied at the tire contact patch producing a 60 Nm transport moment at the hub carrier center (a tire radius of 300 mm is considered).

A 4 Nm moment along Z is applied at the center of the hub carrier. This moment is called self-aligning torque and is produced by the uneven distribution of shear stresses in the tyre/terrain contact patch.

External turn

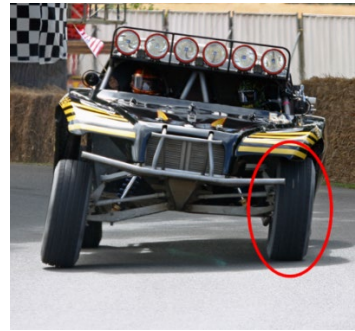
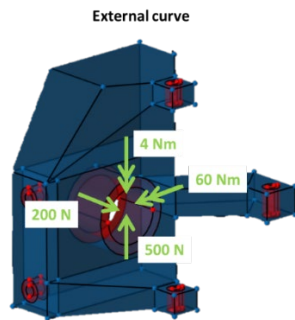


Figure 11 - External turn

Most of the loads are identical to the *Internal turn* load case, but act in opposite directions. The vertical force is now 500 N due to the load transfer.

Hit with an obstacle

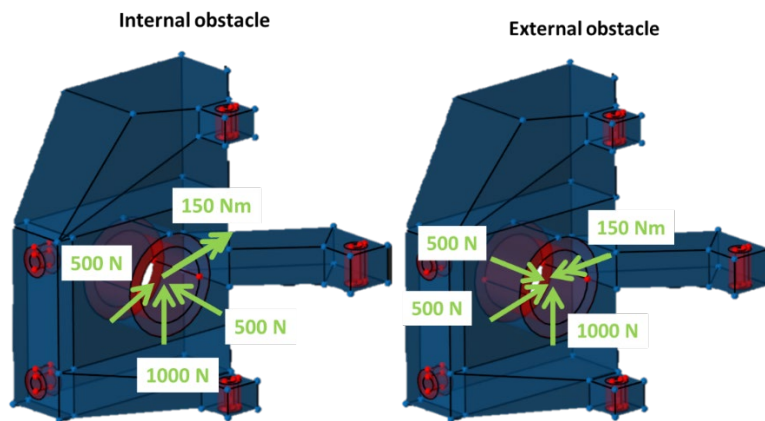


Figure 12 – Internal and external obstacle

These load cases simulate the hitting of the wheel with an obstacle. The obstacle is generally a curb or a hole.

Material

A linear elastic material is considered for the hub carrier. You can assume aluminum 6061 T6 as reference material for the component.

Property	Value
E, elastic modulus	70 GPa
d, density	2700 kg/m ³
Ys, proof strength	270 MPa

Problem definition

Now that loads and support conditions have been defined, the problem can be formulated.

Minimize the weighted compliance for the six load cases under the following stiffness constraints:

Constraint	Value
Camber stiffness	> 160 000 Nm/rad
Toe stiffness	> 84 000 Nm/rad

Where camber stiffness is computed from the rotation of the center of the hub carrier when a unitary moment around X direction is applied at the same point. The toe stiffness is computed from the rotation of the center of the hub carrier when a unitary moment around Z direction is applied at the same point.

In a more compact form, the problem states:

Min	Weighted compliance
Subjected to	Camber stiffness > 160 000 Toe stiffness > 84 000 Mass fraction < threshold (e.g. 0.4)

Manufacturing constraints

It is important to implement manufacturing constraints in order to obtain a structure which can be actually produced. The following two manufacturing technologies can be considered:

- Extrusion along Y direction
- Draw (forging) along Y direction

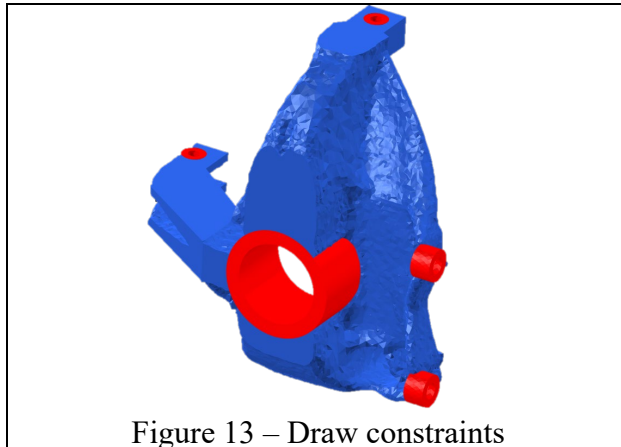


Figure 13 – Draw constraints

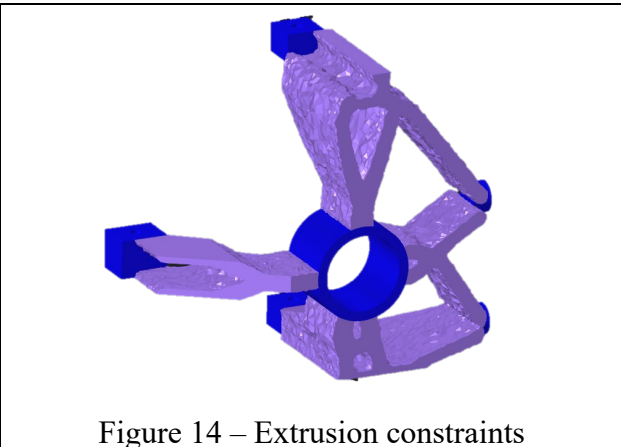


Figure 14 – Extrusion constraints

Requests

Please write a report in which you show:

- The final structure you have obtained, along with its mass and stiffness (camber and toe)
- A finite element analysis showing the displacement and stress field on the component.

Provided material

A hyperworks file (.hm) containing the geometry of the design and fixed domains is given (it can be downloaded from the course homepage in Webeep).